

17-7 PH Stainless Steel, CH900, plate, sheet, and strip

Categories: [Metal](#); [Ferrous Metal](#); [Stainless Steel](#); [Precipitation Hardening Stainless](#); [T S10000 Series Stainless Steel](#)

Material Notes: **Processing:** CH900 - heated to austenitic range, 1040°C (1900°F), and water quenched. Severely cold worked (60-70%). The cold working induces a martensite transformation in the austenite. Tempered at 480°C (900°)

Applications: high strength high temperature applications, chemical processing equipment, heat exchangers, power boilers, superheater tubes

Corrosion Resistance: 17-7 PH is suitable for use in fresh water, industrial and marine atmospheres, and mild chemical and oxidizing environments. 17-7 PH should not be used in salt water or reducing environments.

Key Words: UNS S17700, double treatment alloy, CH900, plate, sheet, and strip, 17-7PH, 17-7 PH, 17/7PH, 17/7 PH, Precipitation Hardening

Vendors: [Click here](#) to view all available suppliers for this material.

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Physical Properties	Metric	English	Comments
Density	7.80 g/cc	0.282 lb/in ³	

Mechanical Properties	Metric	English	Comments
Hardness, Rockwell C	46	46	
Tensile Strength, Ultimate	1650 MPa	239000 psi	
Tensile Strength, Yield	1590 MPa	231000 psi	
Elongation at Break	1.0 %	1.0 %	
Modulus of Elasticity	204 GPa	29600 ksi	

Electrical Properties	Metric	English	Comments
Electrical Resistivity	0.0000830 ohm-cm	0.0000830 ohm-cm	

Thermal Properties	Metric	English	Comments
CTE, linear 	11.0 µm/m-°C @Temperature 0.000 - 100 °C	6.11 µin/in-°F @Temperature 32.0 - 212 °F	
	11.6 µm/m-°C @Temperature 0.000 - 315 °C	6.44 µin/in-°F @Temperature 32.0 - 599 °F	
Specific Heat Capacity	0.460 J/g-°C @Temperature 0.000 - 100 °C	0.110 BTU/lb-°F @Temperature 32.0 - 212 °F	
Thermal Conductivity 	16.4 W/m-K @Temperature 100 °C	114 BTU-in/hr-ft ² -°F @Temperature 212 °F	
	21.8 W/m-K @Temperature 500 °C	151 BTU-in/hr-ft ² -°F @Temperature 932 °F	
Melting Point	1400 - 1450 °C	2550 - 2640 °F	
Solidus	1400 °C	2550 °F	
Liquidus	1450 °C	2640 °F	

Component Elements Properties	Metric	English	Comments
Aluminum, Al	0.75 - 1.5 %	0.75 - 1.5 %	
Carbon, C	<= 0.090 %	<= 0.090 %	
Chromium, Cr	16 - 18 %	16 - 18 %	
Iron, Fe	70.6 - 76.8 %	70.6 - 76.8 %	as balance
Manganese, Mn	<= 1.0 %	<= 1.0 %	
Nickel, Ni	6.5 - 7.75 %	6.5 - 7.75 %	
Phosphorus, P	<= 0.040 %	<= 0.040 %	
Silicon, Si	<= 1.0 %	<= 1.0 %	
Sulfur, S	<= 0.040 %	<= 0.040 %	

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